

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Python, Scala, Kafka, Airflow, Docker, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered Python/Scala **data pipelines** ingesting **50M+ daily events** via Kafka streams and Airflow DAGs, enforcing RBAC/encryption on **petabyte-scale Delta Lake**.
- Optimized **Apache Spark** batch jobs via Spark-native API migration and AQE tuning, cutting latency by **40%**; built dashboards tracking pipeline health and cost.
- Shipping multi-region services on **Kubernetes**-orchestrated, Docker-containerized clusters with fault tolerance, idempotent re-runs, and **Terraform**-provisioned failover for high-reliability workloads.

Software Engineer Intern | *Python, TypeScript/Node.js, AWS, Docker*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Built async **Python** data pipelines on **AWS** (SQS, S3, VPC) processing **thousands of jobs/day**, designing queue topologies with dead-letter routing and back-pressure controls.
- Authored **automated test** suites (E2E + integration) across Docker-containerized stacks, gating all merges and deploys through CI/CD with confidence.
- Raised service reliability to **99.9%** via retry/backoff, **Elasticsearch**-indexed logs, CloudWatch monitoring/alerting, and structured observability across networked services.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *C++, Python, Unreal Engine*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led development of a real-time VR simulation in **C++**, designing an **edge-to-cloud data collection** pipeline streaming sensor events to AI decision-making.
- Built a multithreaded supervisory system (C++/Python) coordinating concurrent actors with shared state, lock-free message passing, and real-time monitoring.
- Profiled the processing pipeline, identified memory and I/O hotspots, and restructured the call graph — achieving **40%** latency reduction and **90%** cost savings.

PROJECTS

Backer | *Go, Python, AWS (DynamoDB, S3), XGBoost*

Hack The Coast 2026

- Designed a **DynamoDB**-backed data layer (10+ tables) with batch operations, transactional writes, and typed repository abstractions supporting high-throughput workflows.
- Built a **Go** data ingestion service processing DynamoDB event streams; engineered features and trained an XGBoost ranking model scoring entities on 6 signals.
- Integrated S3 storage, async notifications, and middleware with role-based access control.

Workbase | *TypeScript, PostgreSQL, AWS Bedrock, Vitest*

Personal Project

- Built a career content platform that ingests GitHub repositories via OAuth, normalizes commit and PR data into **PostgreSQL**, and generates verified resume highlights with evidence attribution.
- Designed a multi-step **data pipeline** — LLM generation, schema validation, structured output repair, and audit logging — with deterministic retry and automated test suites covering pipeline correctness.

SKILLS

Languages: C/C++, Python, Go, Scala, Java, SQL, TypeScript/JavaScript

Infrastructure: Kubernetes, Docker, Terraform, Kafka, Airflow, Apache Spark, AWS (S3, SQS, Lambda, DynamoDB, CloudWatch); Linux, Git, CI/CD

Databases & Storage: PostgreSQL, DynamoDB, Elasticsearch, Redis, Delta Lake; REST APIs, Flask/FastAPI